

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS – JANUARY 2026

Program	: B.E. – Common to all Programs	Semester	: V
Course Name	: Research Methodology and Intellectual Property Rights	Max. Marks	: 100
Course Code	: 21 / 22 / 23 / AL58	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Classify and explain with suitable examples the different types of research. CO1 (10)
 - Analyze how creative and critical thinking contribute to different stages of the research process. CO1 (10)
- Discuss the role of Google scholar by describing the key features that make Google scholar useful for academic research. CO1 (10)
 - Outline the key ethical principles researchers must follow and the consequences of ethical violations in scientific investigations. CO1 (10)

UNIT - II

- What are the different methods of data collection in research? Explain with examples. CO2 (10)
 - Explain the important sampling distributions methods with examples. CO2 (10)
- Explain the meaning of the following in context of research: CO2 (10)
 - Extraneous and confounding variable
 - Independent and dependent variables
 - Experimental and control groups.
 - What is research hypothesis? Explain the types of hypotheses. CO2 (10)

UNIT - III

- What are the different methods of data collection in research? Explain with examples. CO3 (10)
 - The mean of a certain production process is known to be 50 with a standard deviation of 2.5. The production manager may welcome any change in mean value towards higher side but would like to safeguard against decreasing values of mean. He takes a sample of 12 items that gives a mean value of 48.5. What inference should the manager take for the production process on the basis of sample results? Use 5 per cent level of significance for the purpose. CO3 (10)
- What are the characteristics of a good sample design? Describe probability sampling with an example. CO3 (10)
 - A certain process produces 10 per cent defective articles. A supplier of new raw material claims that the use of his material would reduce the proportion of defectives. A random sample of 400 units using this new material was taken out of which 34 were defective units. Can the supplier's claim be accepted? Test at 1 per cent level of significance. CO3 (10)

UNIT- IV

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| 7. | a) | What are the different types of patent applications in India? Briefly discuss the procedure for filing a typical patent in India. | CO4 | (10) |
| | b) | Describe the steps involved in patent search and prior art analysis. Why is it important? | CO4 | (10) |
| 8. | a) | Describe on any four forms of Intellectual Property Rights with an example. | CO4 | (10) |
| | b) | Describe Basmati rice patent case in detail. | CO4 | (10) |

UNIT - V

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| 9. | a) | Enumerate the steps involved in the registration of a design. | CO5 | (10) |
| | b) | Explain in detail the concept of trademarks and outline the step-by-step procedure for their registration in India. | CO5 | (10) |
| 10. | a) | Analyze the issues surrounding trademark infringement by presenting a detailed case study. | CO5 | (10) |
| | b) | Discuss the range of works that are included within the scope of the Copyright Act. | CO5 | (10) |
